

James Yu

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SUMMARY

- Computer Science Master's student specializes in Computer Graphics and Physics-based Simulation.
- Skilled in Ray Tracing, Geometric Processing, Computer Vision, Machine Learning, and Web Development.
- Research experience in High-Performance Computing and Computer-Aided Design.
- Strong background in both hardware and software.

EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science | GPA: 3.71 / 4.0

Atlanta, GA, USA

Expected Dec 2025

National Taiwan University

Bachelor of Science in Electrical Engineering | GPA: 4.14 / 4.30

Taipei, Taiwan

Jun 2022

WORK EXPERIENCE

Teaching Assistant

Computer Vision Course | Georgia Institute of Technology

Atlanta, GA, USA

Aug 2024 - Dec 2024

- Engineered course project frameworks in Python.
- Taught topics such as camera calibration, Stereo, motion detection, and image classification to over 300 students.

Teaching Assistant

Cornerstone EECS Design and Implementation Course | National Taiwan University

Taipei, Taiwan

Mar - Jun, 2021, 2022

- Created assignments regarding graph searching in Python and signal processing through Arduino.
- Developed the online grading platform written in Python and JavaScript.
- Led four teams to achieve full score on the designated project, breaking the all-time course record.

RESEARCH EXPERIENCE

In-Memory Computing on Genome Sequencing

Academia Sinica | Advisor: **Hsiang-Yun Cheng**

Taipei, Taiwan

Jul 2021 - Mar 2023

- Implemented the high-performance Compute-In-Memory simulator for genome sequencing.
- Achieved 40 times faster than the latest in-memory computing genome sequencing tool.

Functional Engineering Change Order with Behavioral Change Guidance

National Taiwan University | Advisor: **Iris Hui-Ru Jiang**

Taipei, Taiwan

Mar 2021 - Sep 2021

- Utilized the Computer-Aided Design to generate patches converting existing circuits to match new specifications.
- Led a team of three novices to win an **Honorable Mention** in 2021 CAD Contest @ ICCAD.

PROJECTS

Fluid Manipulation via Hand Motions

Feb 2024 - Apr 2024

- Detected and identified hand gestures by Computer Vision techniques and machine learning methods.
- Achieved 98% accuracy in gesture detection and real-time simulation with inference time 3.51 ms in average, which is 12 times faster than the existing tool MediaPipe.

Physics-based Simulation of Gas Particles

Nov 2018 - Jan 2019

- Developed a Python program to simulate the behavior of ideal gas particles within a confined space.
- Achieved real-time simulation on over 1000 particles by using optimized Python libraries.
- Achieved 95% in accuracy verified by Ideal Gas Law.

Distribution Ray Tracer

Jan 2024 - Mar 2024

- Designed a high-performance CPU ray tracer in Java featuring Bounding Volume Hierarchy (BVH) acceleration, Instancing, Glossy Material, Area Light, Soft Shadows, Motion Blur, and Depth of Field.
- Achieved 15 times faster ray tracing on a mesh consisting of thousands of triangles by utilizing the BVH acceleration structure.

Disaster Tweets Detection

Oct 2023 - Dec 2023

- Designed a machine learning model to detect the twitter posts which are related to disasters.
- Utilized **Embedding and Vectorizing** models to enhance the language processing.
- Achieved higher than 80% in both precision and recall on a dataset of 1000 tweets.

SKILLS

Programming

Python, C++, RISC-V, Verilog, MATLAB, Arduino, STM32, HTML/CSS, Java, JavaScript, LaTeX, SQL

Expertise

Computer Graphics, Machine Learning, Computer Vision, Modeling and Simulation, Database, Data Visualization, Computer-Aided Design, Computer Architecture, High-Performance Computing

Leisure Activity

Philosophy Thinking, Badminton, Marathon, Frisbee